HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **Python** backend services for agent orchestration, integrating with **LangChain** and **LangGraph** for complex task automation.
- Developed and deployed **custom MCP integrations** to extend agent tool capabilities, enabling real-time interaction with external systems.
- Constructed **RAG pipelines** that provided contextual memory and enterprise data access to AI agents, improving response relevance by approximately **30%**.
- Iterated rapidly on new agent capabilities, prototyping in the morning and hardening for production in the afternoon, shipping an average of **2 new features** per week.
- Containerized agent services using **Docker** and deployed them to a cloud environment, ensuring scalability and efficient resource management.
- Collaborated with cross-functional teams to define technical requirements and estimate sprint efforts for agent development initiatives.
- Wrote and maintained comprehensive **production runbooks** and incident response documentation for agent workflows, reducing average incident resolution time by **15%**.

SKILLS

Agentic AI & Backend – Python, C#, .NET, MCP (Model Context Protocol), Agent Orchestration, Tool-Calling

AI Frameworks & Libraries – LangGraph, LangChain, RAG Architectures, Embeddings, Vector Databases

Backend Development – REST APIs, Microservices, System Design, API Integrations

Cloud & DevOps – Docker, Kubernetes, AWS, CI/CD, Deployment

Data & Analytics – SQL, Data Pipelines, ETL

Methodologies – Agile, Scrum, Rapid Prototyping

PROJECTS

Autonomous Workflow Orchestration Engine

- Engineered a **Python**-based orchestration engine using **LangGraph** to manage multi-step autonomous agent workflows for enterprise task automation.
- Integrated **custom tool-calling capabilities** via **MCP**, allowing agents to interact with internal APIs and external services for data retrieval and processing.
- Implemented comprehensive state management and error handling for long-running agent processes, achieving a **95% success rate** for core automation tasks.

Contextual Memory for AI Agents

- Developed a **RAG pipeline** to provide AI agents with access to a knowledge base derived from enterprise documents and databases.
- Built **Python** data ingestion scripts to process and index unstructured data into a **vector database** for efficient retrieval by the RAG system.
- Designed and implemented retrieval strategies that improved agent response accuracy by approximately **25%** in simulated user interaction tests.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)